CYBERSECURITY HOMELAB DOCUMENTATION

Network Security and SOC Monitoring Platform

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1. Executive Summary

Project Overview

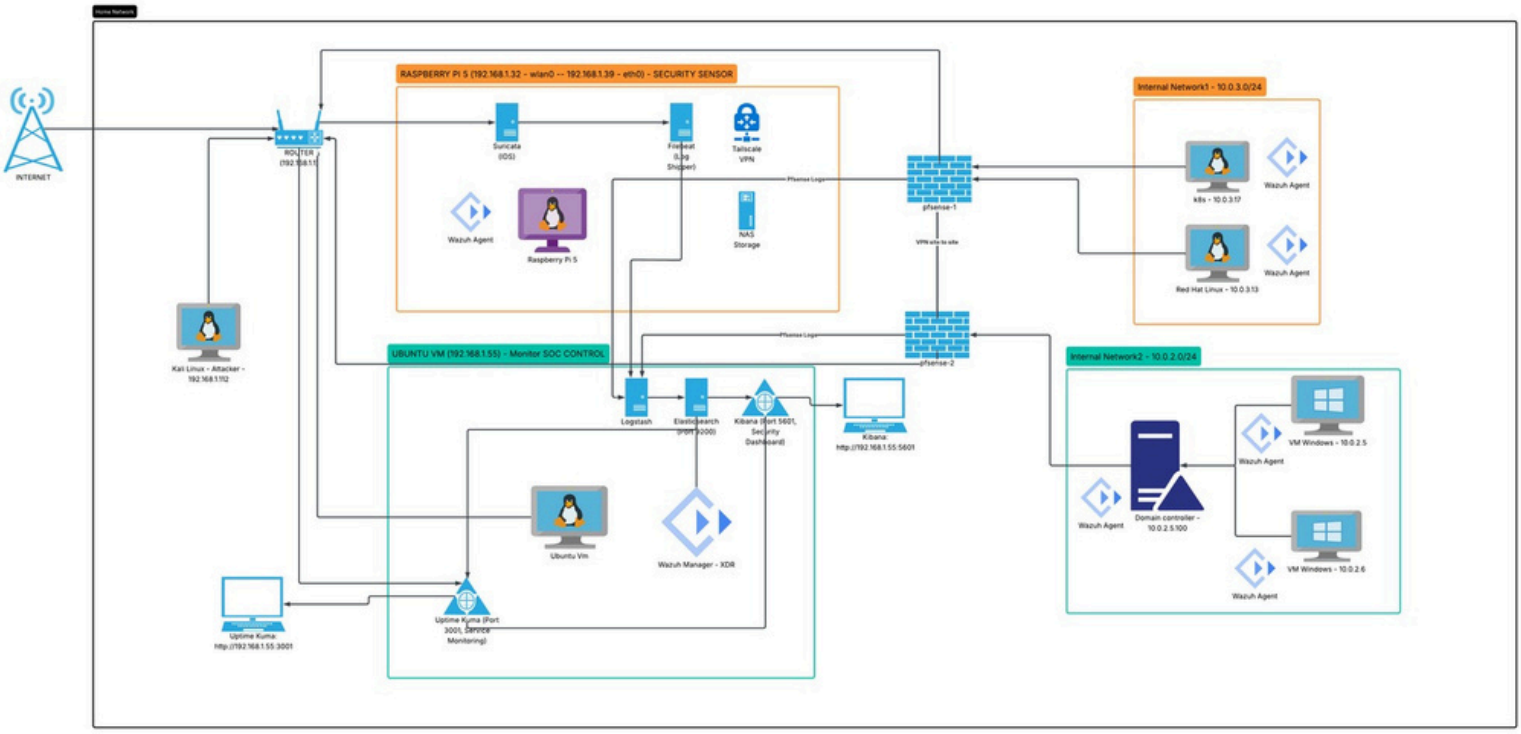
Thisdocumentdescribes the complete implementation of an enterprise-grade Security Operations Center (SOC) homelab. The platform provides centralized security monitoring across a multi-site network environment, demonstrating professional security capabilities including:

- Network Security: Dual pfSense firewalls with site-to-site VPN
- Intrusion Detection: Suricata IDS/IPS with 65,000+ detection rules
- Endpoint Security: Wazuh XDR across Windows and Linux endpoints
- Centralized Logging: Complete ELK Stack (Elasticsearch, Logstash, Kibana)
- Service Monitoring: Uptime Kuma for infrastructure health
- Attack Simulation: Kali Linux for detection validation

Key Achievements

Capability	Implementation
Multi-site firewall monitoring	2x pfSense with site-to-site VPN
Network traffic analysis	Suricata on dedicated Raspberry Pi 5
Endpoint detection	Wazuh agents on 5+ endpoints
Centralized log management	ELK Stack processing 10,000+ logs/hour
Real-time dashboards	3 professional Kibana dashboards
Attack validation	5+ attack types simulated and detected

2. Architecture Overview



192.168.1.32

```
cs-cod3x@raspberrypi:~ $ docker ps
CONTAINER ID   IMAGE                                   COMMAND                  CREATED        STATUS        PORTS     NAMES
f9a249a99b70   docker.elastic.co/beats/filebeat:7.17.15   "/usr/bin/tini -- /u…"   24 hours ago   Up 22 hours             filebeat
70c593badd69   jasonish/suricata:latest                "/docker-entrypoint.…"   30 hours ago   Up 30 hours             suricata
cda2759933b3   jasonish/suricata:latest                "/docker-entrypoint.…"   11 days ago    Up 4 days               suricata-sensor
cs-cod3x@raspberrypi:~ $
```

```
CS-COD3X@CS-COD3X:~$ docker ps
CONTAINER ID   IMAGE                                                   COMMAND                  CREATED        STATUS                 PORTS
NAMES
f41a2cec19cb   docker.elastic.co/logstash/logstash:7.17.15             "/usr/local/bin/dock…"   22 hours ago   Up 22 hours            5044/tcp, 9600/tcp, 0.0.
0.0:5140->5140/tcp, 0.0.0.0:5140->5140/udp, :::5140->5140/tcp, :::5140->5140/udp
logstash
710f61fd40c7   docker.elastic.co/kibana/kibana:7.17.15                 "/bin/tini -- /usr/l…"   42 hours ago   Up 30 hours            0.0.0.0:5601->5601/tcp,
:::5601->5601/tcp
kibana
3be7d931a308   docker.elastic.co/elasticsearch/elasticsearch:7.17.15   "/bin/tini -- /usr/l…"   43 hours ago   Up 30 hours            0.0.0.0:9200->9200/tcp,
:::9200->9200/tcp, 9300/tcp
elasticsearch
83e0734aa1d7   wazuh/wazuh-manager:4.9.0                               "/init"                  4 days ago     Up 22 seconds          0.0.0.0:1514-1516->1514-
1516/tcp, :::1514-1516->1514-1516/tcp, 514/udp, 0.0.0.0:55000->55000/tcp, :::55000->55000/tcp
wazuh-manager_wazuh.manager_1
82ba98abf2c2   grafana/grafana:latest                                  "/run.sh"                4 days ago     Up 30 hours            0.0.0.0:3000->3000/tcp,
:::3000->3000/tcp
grafana
54da74bf7a9b   louislam/uptime-kuma:latest                             "/usr/bin/dumb-init …"   8 days ago     Up 30 hours (healthy)  0.0.0.0:3001->3001/tcp,
:::3001->3001/tcp
uptime-kuma
5b1e8d5d34f1   wordpress:latest                                        "docker-entrypoint.s…"   18 months ago  Up 30 hours            0.0.0.0:8000->80/tcp, [:
:]:8000->80/tcp
wordpress_wordpress_1
b807cfd41ec7   mysql:5.7                                               "docker-entrypoint.s…"   18 months ago  Up 30 hours            3306/tcp, 33060/tcp
wordpress_db_1
```

Security Zones

Zone	Network	Purpose	Access Control
Management	192.168.1.0/24	SOC tools, firewall management	MANAGE_NETS alias
Site A Internal	10.0.2.0/24	Windows, DC, Linux workloads	pfSense firewall rules
Site B Internal	10.0.3.0/24	K8s, Red Hat, Kali attacker	pfSense firewall rules
WAN	192.168.1.x/24	External connectivity	Default deny

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3. Infrastructure Components

Hardware Requirements

Device	Specs	Role
Raspberry Pi 5	8GB RAM, 128GB SD	Security Sensor (Suricata)
Ubuntu Server VM	4 vCPU, 8GB RAM, 100GB	SOC Controller
pfSense-1 VM	2 vCPU, 2GB RAM	Site A Firewall
pfSense-2 VM	2 vCPU, 2GB RAM	Site B Firewall
Windows VM	2 vCPU, 4GB RAM	Endpoint
Kali Linux VM	2 vCPU, 4GB RAM	Attack Simulation

Software Stack

Component	Technology	Version	Purpose
Firewall	pfSense CE	2.7.2	Network security & VPN
IDS/IPS	Suricata	7.0.5	Network traffic analysis
SIEM	Wazuh/ELK	4.7.0	Endpoint security & XDR
Log Shipping	Filebeat	7.17.15	Log aggregation
Log Processing	Logstash	7.17.15	Log parsing & enrichment Log storage & indexing
Search & Analytics	Elasticsearch	7.17.15	Dashboards & alerts
Visualization	Kibana	7.17.15	
Monitoring	Uptime Kuma	1.23.0	Service health checks
VPN	Tailscale	Latest	Secure remote access

Service Endpoints

Service	Port	URL	Purpose
Kibana	5601	http://192.168.1.55:5601	Security Dashboards

Uptime Kuma	3001	http://192.168.1.55:3001	Service Monitoring
pfSense-1 GUI	8443	https://192.168.1.40:8443	Site A Firewall
pfSense-2 GUI	8443	https://192.168.1.41:8443	Site B Firewall
Elasticsearch API	9200	http://192.168.1.55:9200	Log queries
Wazuh Manager	1514/1515	TCP only	Agent communication

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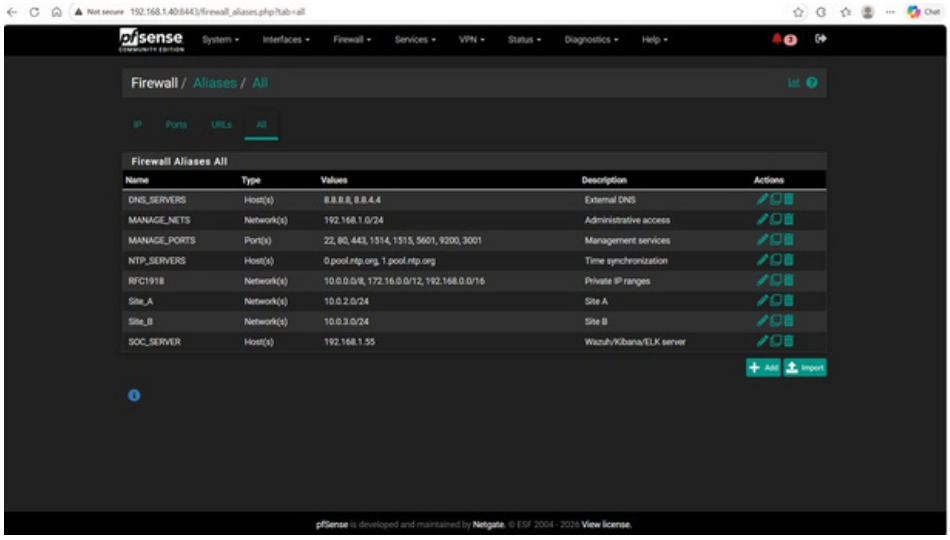
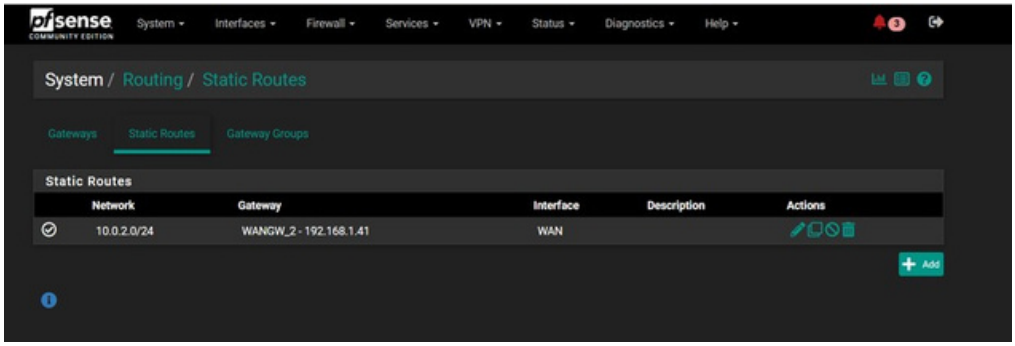
4. Network Configuration

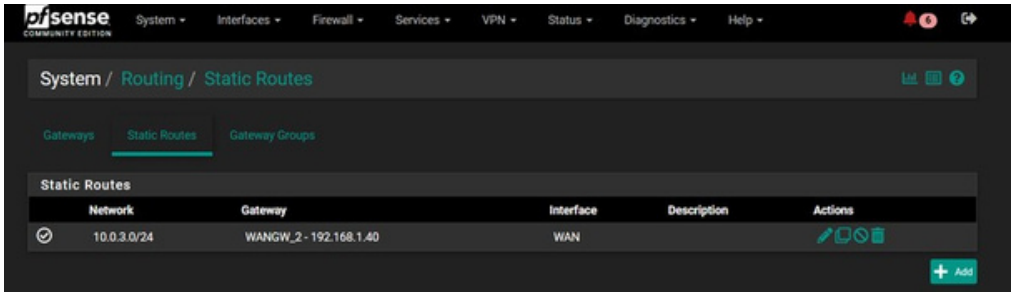
pfSense Aliases



Static Routes



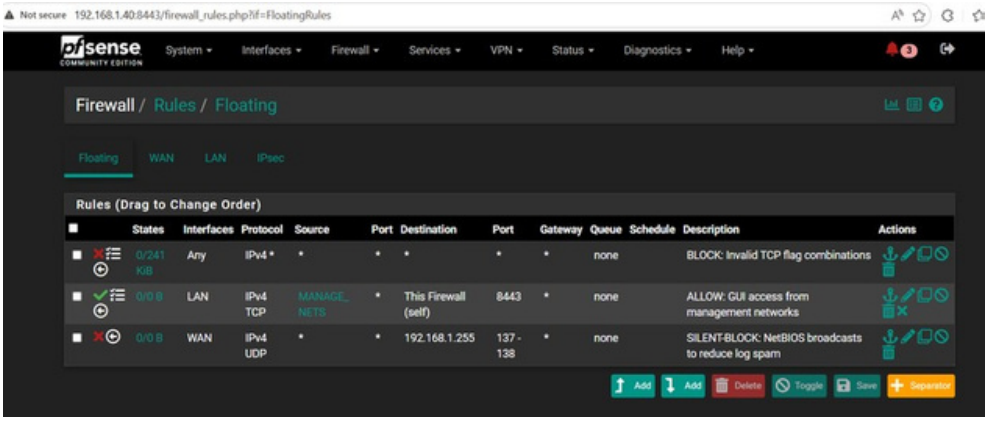
Pfsense-1



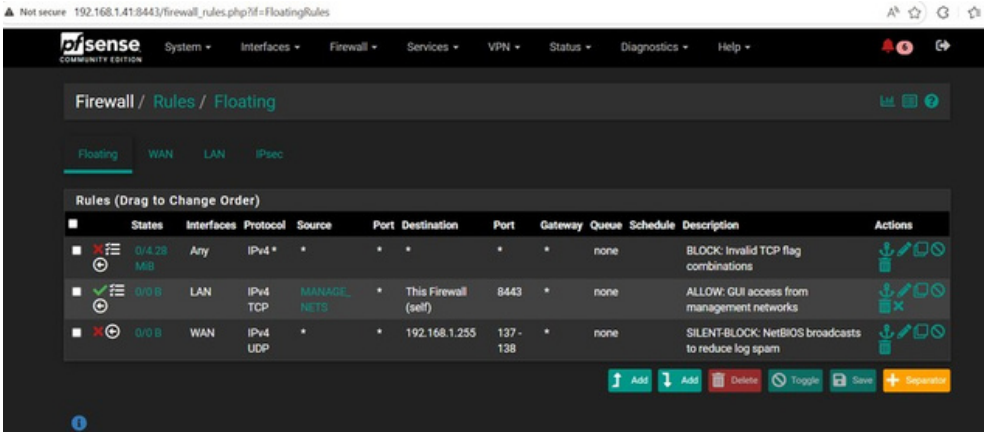
Pfsense-2

5. pfSense Firewall Configuration

Floating Rules

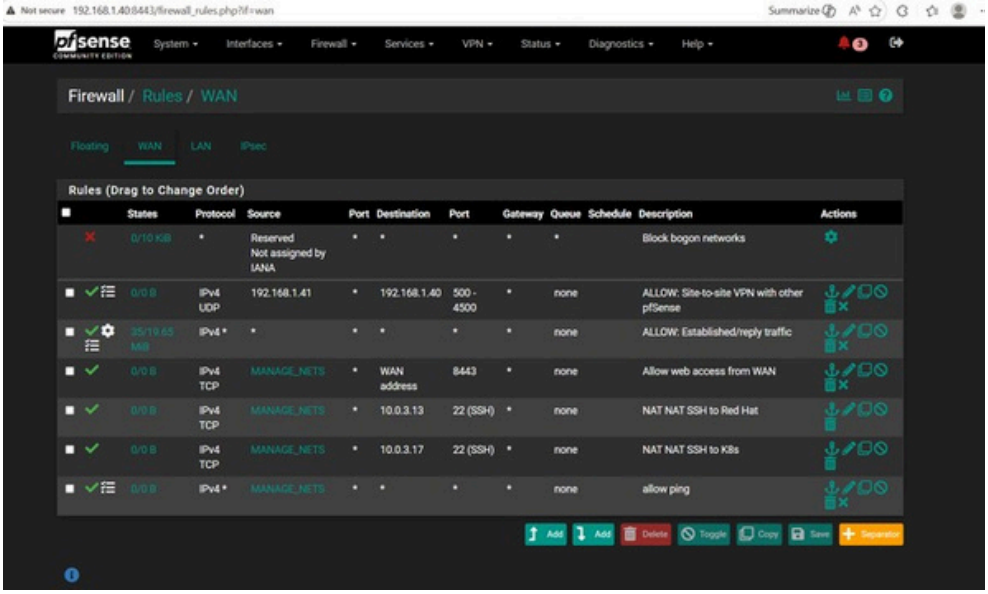


Pfsense-1

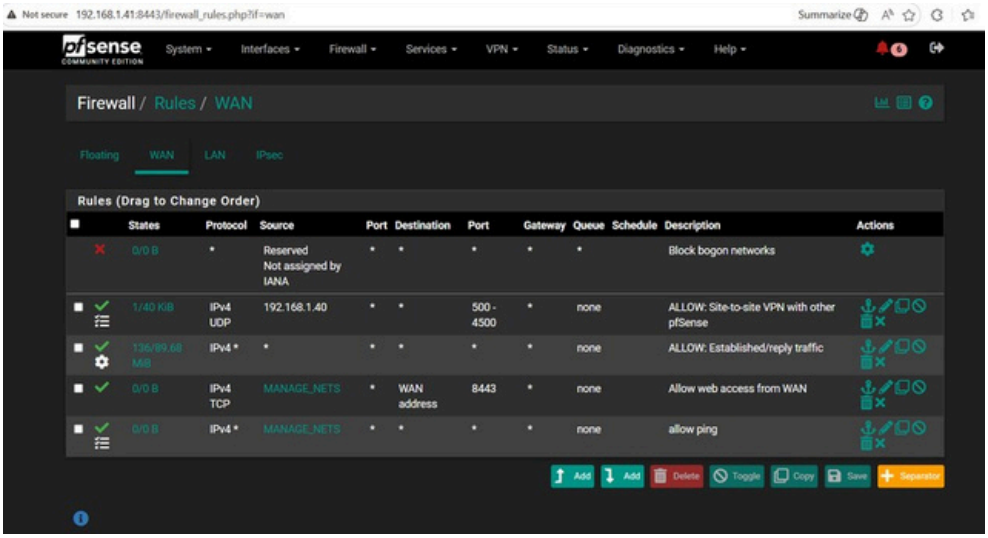


Pfsense-2

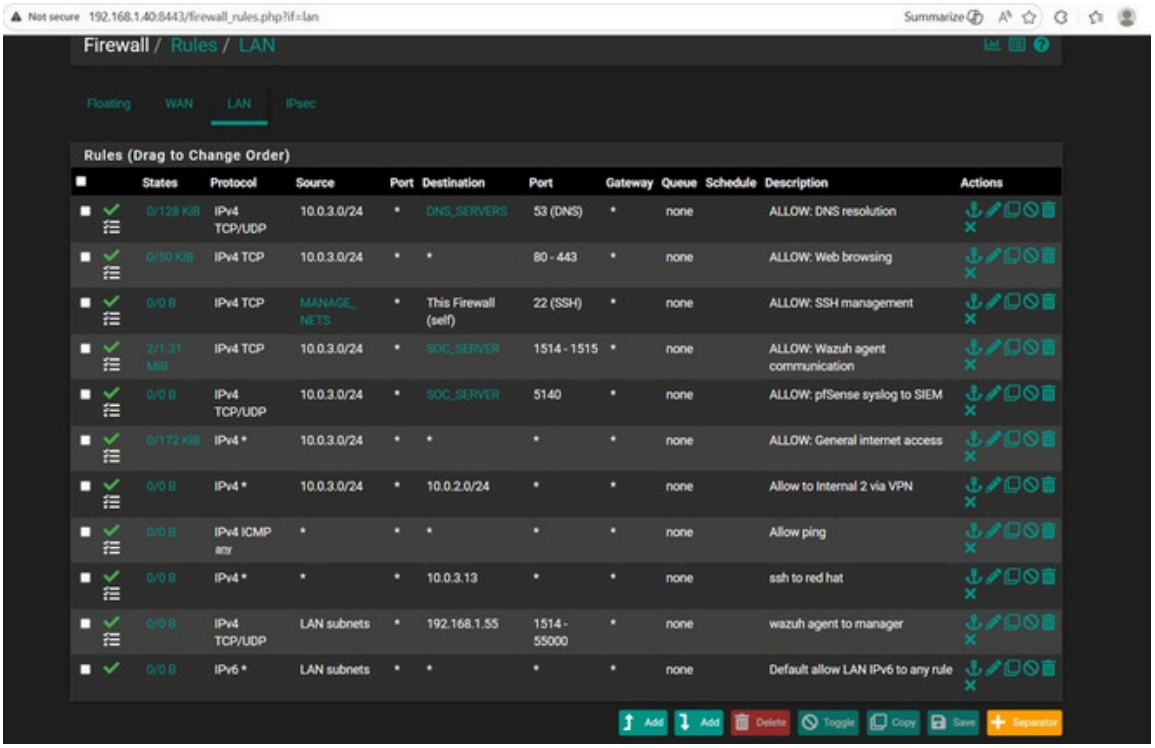
WAN Rules (pfSense-1 - Site B)



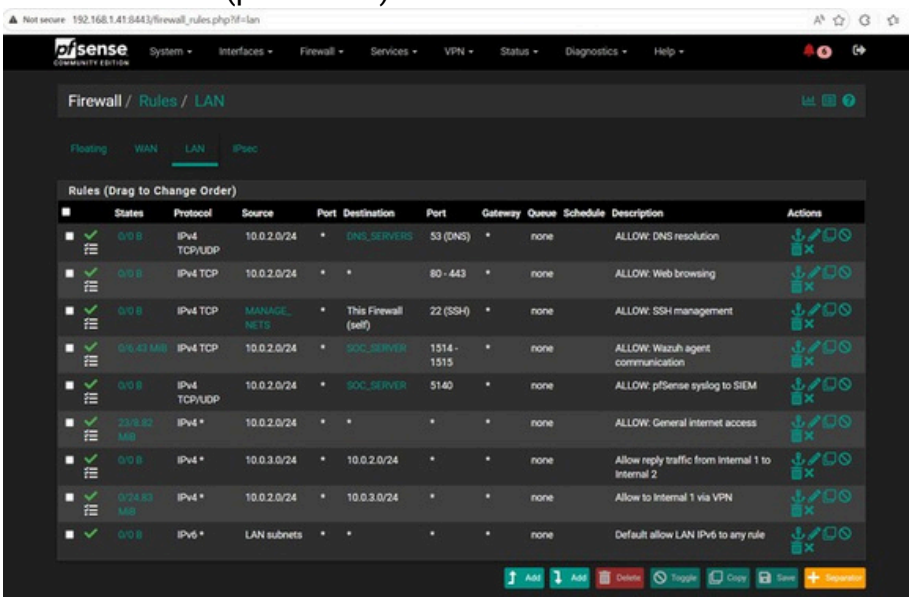
WAN Rules (pfSense-2 - Site A)



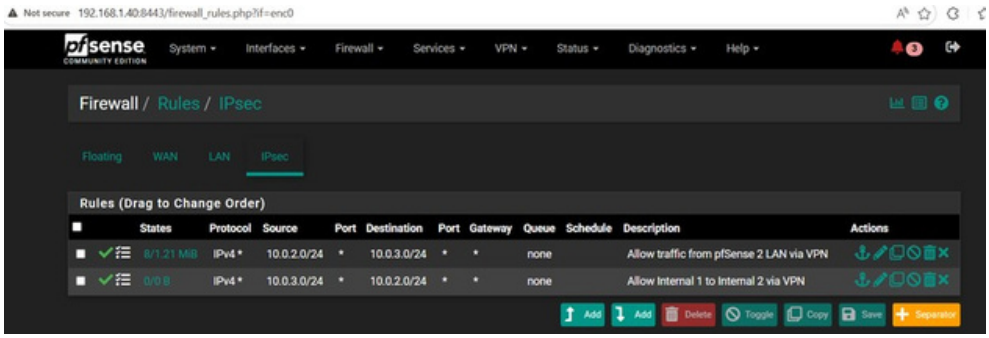
LAN Rules (pfSense-1)

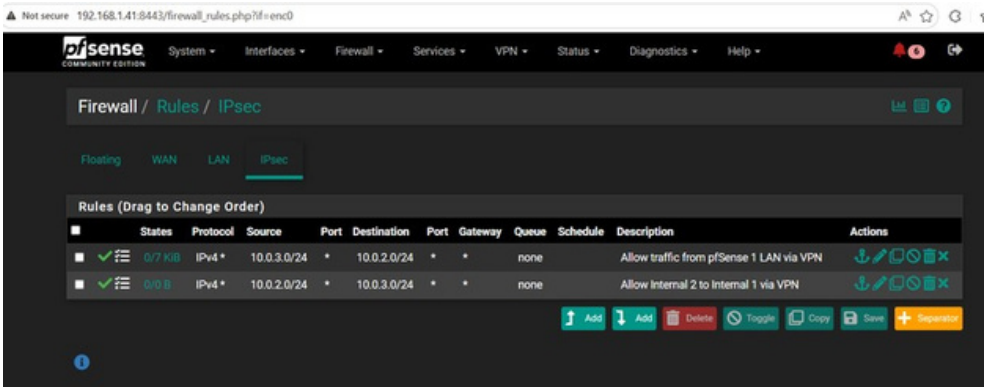


LAN Rules (pfSense-2)

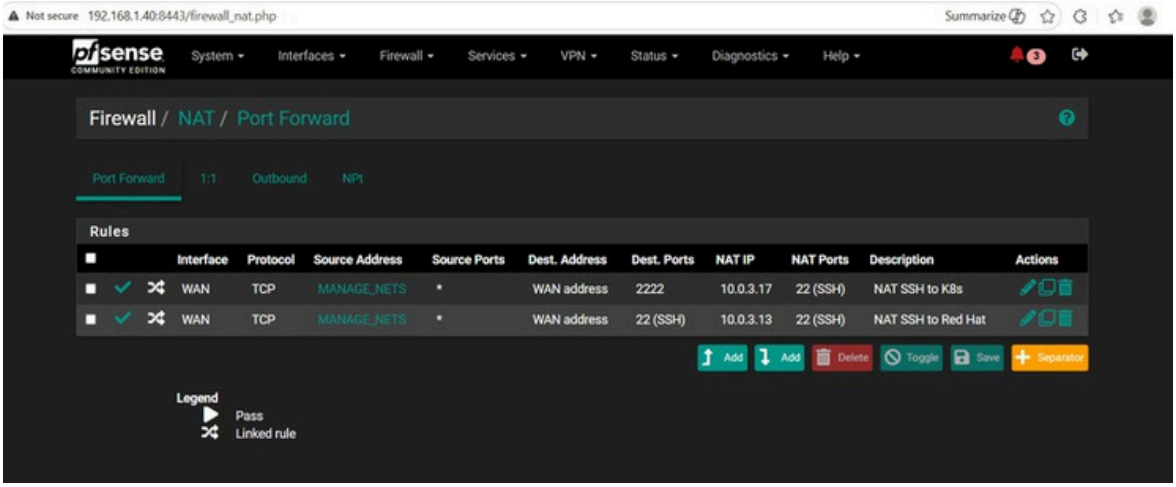


IPsec Rules (Both Firewalls)

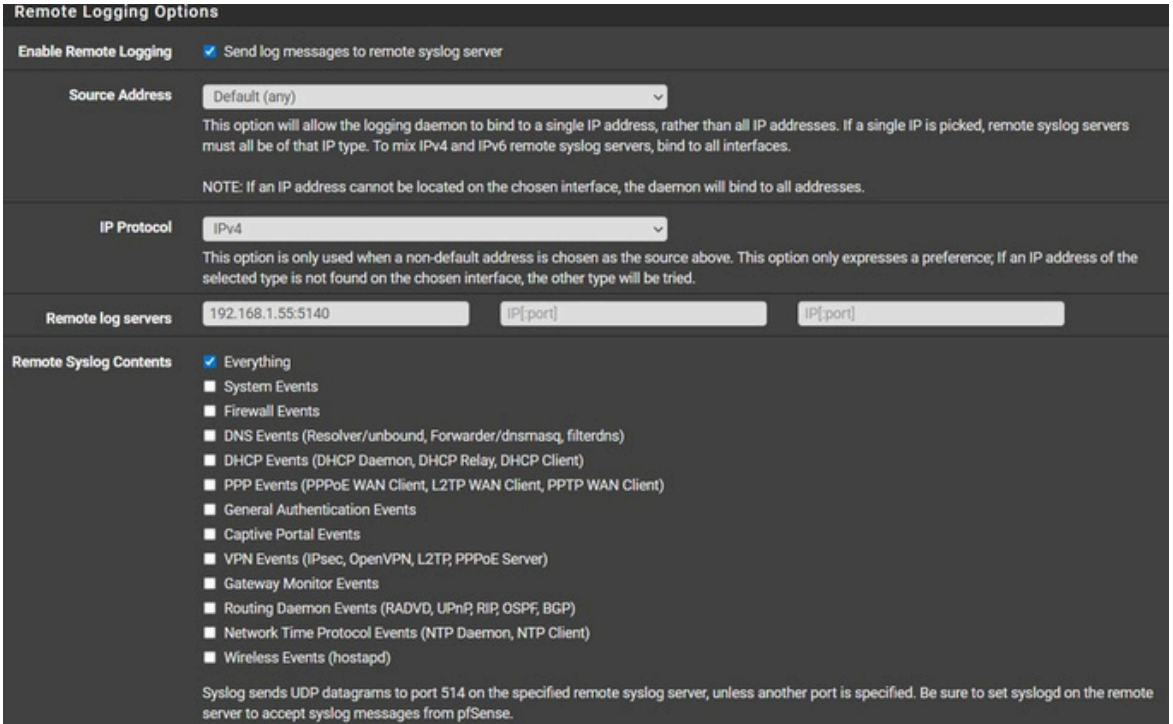




NAT PortForward (pfSense-1Only)



Remote syslog



6. Suricata IDS/IPS Configuration

Deployment

```
# Raspberry Pi deployment
docker run -d \
  --name suricata \
  --restart always \
  --network host \
  --cap-add=NET_ADMIN \
  --cap-add=NET_RAW \
  -v /var/log/suricata:/var/log/suricata \
  jasonish/suricata:latest \
  -i eth0
```

Rule Statistics

Metric	Value
Total rules loaded	65,796
Alert rules	49,920
Enabled rules	65,796
Update frequency	Daily

Key Alert Categories

Category	Example Rules
Malware C2	ET MALWARE C2 Channel
Port Scanning	ET SCAN Nmap
Exploit Detection	ET EXPLOIT EternalBlue
Suspicious Protocols	ET INFO Suspicious User-Agent
Web Attacks	SQL injection, XSS

7. Wazuh XDR Configuration

Wazuh Manager (Ubuntu VM)

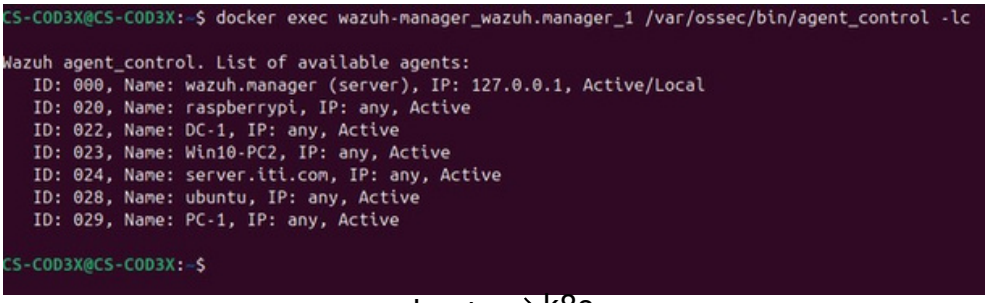
```
# docker-compose.yml (simplified)
services:
wazuh.manager:
    image: wazuh/wazuh-manager:4.7.0
    ports:
      - "1514:1514"
      - "1515:1515"
    volumes:
      - ./config:/var/ossec/etc
```

Enrolled Agents

Agent	OS	Purpose
Windows VM	Windows 10	Endpoint monitoring
Windows DC	Windows Server	Domain controller
Red Hat	RHEL 8	Production server
K8s Node	Linux	Container host
Ubuntu Sensor	Ubuntu 20.04	Additional monitoring

Active Decoders

- Windows Event Channel
- Linux auditd
- File Integrity Monitoring (FIM)
- Malware detection signatures
- Rootkit detection



ubuntu → k8s
server.iti.com → red hat

8. ELK Stack Configuration

Logstash Configuration (pfSense Logs)

```
input {
 tcp { port => 5140 codec => line }
udp { port => 5140 codec => line }
}

filter {
  if[message]=~"filterlog" {
    csv{
      source => "message"
      separator => ","
      columns=>["full_message", "col2", "col3", "if_name", "reason",
                  "action", "direction", "ip_version", "tos", "ecn", "ttl",
                  "length", "id", "offset", "flags", "protocol_id", "protocol",
                  "src_port", "src_ip", "dest_ip", "dest_port", "data_len"]
    }
  }
}

output {
elasticsearch {
    hosts => ["elasticsearch:9200"]
    index => "pfsense-%{+YYYY.MM.dd}"
  }
}
```

Filebeat Configuration (Suricata Logs)

```
filebeat.inputs:
- type: log
enabled: true
paths:
    - /var/log/suricata/eve.json
 json.keys_under_root: true
json.overwrite_keys: true

output.elasticsearch:
 hosts: ["192.168.1.55:9200"]
index: "suricata-%{+yyyy.MM.dd}"

setup.template.name: "suricata"
setup.template.pattern: "suricata-*"
setup.ilm.enabled: false
```

Elasticsearch Indices

Index Pattern	Document Count	Purpose
pfsense-*	2,000+	Firewall logs
suricata-*	700+	IDS alerts
wazuh-alerts-*	500+	Endpoint alerts
wazuh-archives-*	10,000+	Raw agent logs

```
CS-C0D3X@CS-C0D3X:~$ curl -s "http://192.168.1.55:9200/_cat/indices?v" | tee es_indices.txt
health status index                            uuid                   pri rep docs.count docs.deleted store.size pri.store.size
yellow open   wazuh-logs-2026.04.29            LLbQFslaRoKTQ6bmQCCzyQ   1   1      56890            0     25.1mb         25.1mb
yellow open   wazuh-alerts-2026.04.30          6MIEDZjFSWW6nQHC-uXGqw   1   1       9156            0     12.4mb         12.4mb
green  open   .apm-agent-configuration         8DVFqwMoRqqN0Y5GrBxaOw   1   0          0            0       227b           227b
yellow open   wazuh-alerts-2026.05.01          0i8z2PfKRpSaplHZOibX3Q   1   1       5903            0      7.7mb          7.7mb
yellow open   suricata-2026.04.30              7AD8OWMjTrauE8t3Czsj3w   1   1     120703            0       59mb           59mb
yellow open   pfsense-2026.05.01               4Mf5bgDRTVaYNr4cDhJdZQ   1   1     143038            0     33.6mb         33.6mb
green  open   .tasks                           ChlaEwpdSuiHdrCwFNVuXQ   1   0          2            0     13.8kb         13.8kb
yellow open   suricata-2026.05.01              acosHF0pSO-aVn9AUzw_Lw   1   1     309896            0    210.6mb        210.6mb
yellow open   pfsense-2026.04.30               xjWSxV3rSluuo0sx2eJkCA   1   1     155874            0     16.4mb         16.4mb
yellow open   wazuh-alerts-2026.04.29          k5etwnBnTaqqKonykCSnug   1   1        474            0    934.3kb        934.3kb
green  open   .geoip_databases                 5FcjefKVQuWR98tjadEMfg   1   0         42            0       40mb           40mb
yellow open   logstash-2026.04.30              chaKipvAShG6j5x4j9jKOg   1   1      10613            0      5.5mb          5.5mb
green  open   .apm-custom-link                 XiN9ifTaRPq1D4rLAZxOYw   1   0          0            0       227b           227b
yellow open   filebeat-2026.04.29              0QwB55Z-RDSwRgdM9dEU6w   1   1       1166            0      1.8mb          1.8mb
green  open   .async-search                    Ypi2qGgoRSOQ6WHzuvcbRQ   1   0          3            0      7.6kb          7.6kb
green  open   .kibana_7.17.15_001              VaCum4DIQ5yC4y_b5Vw0dQ   1   0        386            2      2.4mb          2.4mb
green  open   .kibana_task_manager_7.17.15_001 9MuQ2pobSP6Z-P_JL7afbg   1   0         17         7338      1.4mb          1.4mb
CS-C0D3X@CS-C0D3X:~$
```

```
CS-C0D3X@CS-C0D3X:~$ docker logs logstash --tail 20 2>&1 | tee logstash_logs.txt
[2026-04-30T18:56:15,828][INFO ][logstash.outputs.elasticsearch][main] Elasticsearch pool URLs updated {:changes=>{:removed=>[], :added=>[http://elasticsearch:9
200/]}}
[2026-04-30T18:56:15,849][WARN ][logstash.outputs.elasticsearch][main] Restored connection to ES instance {:url=>"http://elasticsearch:9200/"}
[2026-04-30T18:56:15,859][INFO ][logstash.outputs.elasticsearch][main] Elasticsearch version determined (7.17.15) {:es_version=>7}
[2026-04-30T18:56:15,859][WARN ][logstash.outputs.elasticsearch][main] Detected a 6.x and above cluster: the `type` event field won't be used to determine the d
ocument _type {:es_version=>7}
[2026-04-30T18:56:15,889][INFO ][logstash.outputs.elasticsearch][main] Config is not compliant with data streams. `data_stream => auto` resolved to `false`
[2026-04-30T18:56:15,889][INFO ][logstash.outputs.elasticsearch][main] Config is not compliant with data streams. `data_stream => auto` resolved to `false`
[2026-04-30T18:56:15,891][INFO ][logstash.javapipeline    ][.monitoring-logstash] Starting pipeline {:pipeline_id=>".monitoring-logstash", "pipeline.workers"=>1
, "pipeline.batch.size"=>2, "pipeline.batch.delay"=>50, "pipeline.max_inflight"=>2, "pipeline.sources"=>["monitoring pipeline"], :thread=>"#<Thread:0x374b84f90/
usr/share/logstash/logstash-core/lib/logstash/pipeline_action/create.rb:54 run>"}
[2026-04-30T18:56:15,908][INFO ][logstash.outputs.elasticsearch][main] Using a default mapping template {:es_version=>7, :ecs_compatibility=>:disabled}
[2026-04-30T18:56:15,943][WARN ][deprecation.logstash.filters.grok][main] Relying on default value of `pipeline.ecs_compatibility`, which may change in a future
 major release of Logstash. To avoid unexpected changes when upgrading Logstash, please explicitly declare your desired ECS Compatibility mode.
[2026-04-30T18:56:16,219][INFO ][logstash.javapipeline    ][main] Starting pipeline {:pipeline_id=>"main", "pipeline.workers"=>2, "pipeline.batch.size"=>125, "p
ipeline.batch.delay"=>50, "pipeline.max_inflight"=>250, "pipeline.sources"=>["/usr/share/logstash/pipeline/logstash.conf"], :thread=>"#<Thread:0x37f4f05d run>"}
[2026-04-30T18:56:16,718][INFO ][logstash.javapipeline    ][.monitoring-logstash] Pipeline Java execution initialization time {"seconds"=>0.82}
[2026-04-30T18:56:16,808][INFO ][logstash.javapipeline    ][.monitoring-logstash] Pipeline started {"pipeline.id"=>".monitoring-logstash"}
[2026-04-30T18:56:16,923][INFO ][logstash.javapipeline    ][main] Pipeline Java execution initialization time {"seconds"=>0.7}
[2026-04-30T18:56:17,044][INFO ][logstash.javapipeline    ][main] Pipeline started {"pipeline.id"=>"main"}
[2026-04-30T18:56:17,060][INFO ][logstash.inputs.tcp      ][main][8f60a125ead4916acfbea19ad12f042a02a94613d0667d584267822ac82459d0] Starting tcp input listener
{:address=>"0.0.0.0:5140", :ssl_enable=>false}
[2026-04-30T18:56:17,082][WARN ][deprecation.logstash.codecs.line][main][fa66c9a35a53acc3b0f64e9219d5553feca5475d17d286920720be4e3e303c9d] Relying on default va
lue of `pipeline.ecs_compatibility`, which may change in a future major release of Logstash. To avoid unexpected changes when upgrading Logstash, please explici
tly declare your desired ECS Compatibility mode.
[2026-04-30T18:56:17,096][WARN ][deprecation.logstash.codecs.line][main][fa66c9a35a53acc3b0f64e9219d5553feca5475d17d286920720be4e3e303c9d] Relying on default va
lue of `pipeline.ecs_compatibility`, which may change in a future major release of Logstash. To avoid unexpected changes when upgrading Logstash, please explici
tly declare your desired ECS Compatibility mode.
[2026-04-30T18:56:17,112][INFO ][logstash.inputs.udp      ][main][fa66c9a35a53acc3b0f64e9219d5553feca5475d17d286920720be4e3e303c9d] Starting UDP listener {:addr
ess=>"0.0.0.0:5140"}
[2026-04-30T18:56:17,156][INFO ][logstash.agent           ] Pipelines running {:count=>2, :running_pipelines=>[:".monitoring-logstash", :main], :non_running_pip
elines=>[]}
[2026-04-30T18:56:17,178][INFO ][logstash.inputs.udp      ][main][fa66c9a35a53acc3b0f64e9219d5553feca5475d17d286920720be4e3e303c9d] UDP listener started {:addre
ss=>"0.0.0.0:5140", :receive_buffer_bytes=>"106496", :queue_size=>"2000"}
CS-C0D3X@CS-C0D3X:~$
```

9. Kibana Dashboards

Dashboard 1: Main Security Dashboard

Panels Included:

1. Metric Row
   - Total Alerts
   - Blocked Connections
   - Unique Attacking IPs
   - Critical Alerts
2. Alerts Over Time (Line Chart)
   - Shows trend of security events
   - Breakdown by alert type
3. Top Attack Sources (Table)

   Top 10 source IPs by block count Linked destination ports
4. Protocol Distribution (Pie Chart)
   - TCP vs UDP vs ICMP traffic
5. Most Targeted Ports (Bar Chart)
   - Top 10 destination ports attacked
6. Recent Events (Data Table)
   - Time, source, destination, action, protocol

Dashboard 2: Network Security Dashboard

Panels Included:

1. pfSense Overview
   - Total passed/blocked traffic
   - Interface traffic breakdown
2. Suricata Alerts
   - Alert severity distribution
   - Top alert signatures
3. VPN Health Gauges
   - Site A → Site B tunnel status
   - Site B → Site A tunnel status
4. Geographic Map (optional)
   - Attack origins by country

Dashboard 3: Endpoint Security Dashboard (Wazuh)

Panels Included:

1. Agent Overview

   Active agents count
   Alerts by agent
   Agent status indicators
2. Alerts by MITRE ATT&CK Tactic
   - Horizontal bar chart
   - Tactic breakdown
3. Top Alert Rules
   - Most frequent detections
   - Severity levels
4. Recent Endpoint Alerts
   - Table with agent name, rule description, severity

Not secure 192.168.1.55:5601/app/management/kibana/indexPatterns

elastic

Stack Management > Index patterns

Management

Ingest
Ingest Pipelines

Data
Index Management
Index Lifecycle Policies
Snapshot and Restore
Rollup Jobs
Transforms
Remote Clusters

Alerts and Insights
Rules and Connectors
Reporting
Machine Learning Jobs

Kibana
Index Patterns
Saved Objects
Tags
Search Sessions
Spaces
Advanced Settings

Stack
License Management
Upgrade Assistant

Index patterns

Create index pattern

Create and manage the index patterns that help you retrieve your data from Elasticsearch.

Search...

Pattern
wazuh-alerts-* Default
pfsense-*
pfsense-*,suricata-*
suricata-*
wazuh-archives-*
wazuh-archives-*
zeek-*

Rows per page: 10

1

elastic

Dashboard

Dashboards

Create dashboard

Search...

Tags

Title	Description	Tags	Actions
Endpoint Security Dashboard (Wazuh)			
Network Security Monitoring			

Rows per page: 20

1

elastic

Dashboard > Endpoint Security Dashboard (Wazuh)

Full screen Share Clone Edit

Search KQL Last 15 minutes Show dates Refresh

+ Add filter

Total Alerts (24h)

99

Count of records

Critical Alerts

0

Count of records

Alerts by MITRE ATT&CK Tactic

Count of records

Alerts by Agent/Words

Count of records

elastic

Dashboard > Network Security Monitoring

Full screen Share Clone Edit

Search KQL Last 15 minutes Show dates Refresh

+ Add filter

Allowed vs Blocked Traffic

block

pass

Unique count of src_ip.keyword

@timestamp per 30 seconds

Total Firewall Events

4,293

Count of records

Blocked Connections

Suricata Alerts

Not secure 192.168.1.55:5601/app/discover#/?_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(from:now-15m,to:now))&_a=(columns:!(src_ip,src_port,dest_ip,dest_port,action),...

elastic | Search Elastic

Discover | Options | New | Open | Share | Inspect | Save

Search | KQL | Last 15 minutes | Show dates | Refresh

+ Add filter

pfsense-*

Search field names

Filter by type 0

Selected fields 5
- src_ip
- src_port
- dest_ip
- dest_port
- action

Available fields 19
- _id
- _index
- _score
- _type
- @timestamp
- @version
- column23
- column24
- column25
- column27
- column29
- direction
- host

4,180 hits | Chart options

May 1, 2026 @ 19:53:21.906 - May 1, 2026 @ 20:08:21.906

Time	src_ip	src_port	dest_ip	dest_port	action
May 1, 2026 @ 20:08:02.896	10.0.3.17	64	8.8.8.8	38,554	pass
May 1, 2026 @ 20:08:02.896	10.0.3.17	64	8.8.8.8	38,554	pass
May 1, 2026 @ 20:08:02.589	127.0.0.1	68	127.0.0.1	29,668	pass
May 1, 2026 @ 20:08:02.589	127.0.0.1	68	127.0.0.1	29,668	block
May 1, 2026 @ 20:08:01.893	127.0.0.1	60	127.0.0.1	13,323	block
May 1, 2026 @ 20:08:01.893	127.0.0.1	60	127.0.0.1	10,296	block
May 1, 2026 @ 20:08:01.591	10.0.2.100	60	202.12.27.33	62,497	pass
May 1, 2026 @ 20:08:01.591	10.0.2.1	68	10.0.2.100	25,209	pass
May 1, 2026 @ 20:08:01.590	10.0.2.100	68	198.97.190.53	63,373	pass
May 1, 2026 @ 20:08:01.590	53	50,857	58	-	block
May 1, 2026 @ 20:08:01.590	10.0.2.100	60	202.12.27.33	63,121	pass
May 1, 2026 @ 20:08:00.591	10.0.2.100	68	192.58.128.30	62,689	pass
May 1, 2026 @ 20:07:59.590	53	50,857	58	-	block

elastic | Search Elastic

Discover | Options | New | Open | Share | Inspect | Save

Search | KQL | Last 15 minutes | Show dates | Refresh

+ Add filter

suricata-*

Search field names

Filter by type 0

Selected fields 5
- src_ip
- src_port
- dest_ip
- dest_port
- flow.alerted

Available fields 583

Popular
- alert.action
- alert.category
- alert.signature

- _id
- _index
- _score
- _type
- @timestamp
- agent.ephemeral_id
- agent.hostname
- agent.id

4,056 hits | Chart options

May 1, 2026 @ 19:54:27.713 - May 1, 2026 @ 20:09:27.713

Time	src_ip	src_port	dest_ip	dest_port	flow.alerted
May 1, 2026 @ 20:09:20.556	192.168.1.37	56,895	239.255.255.250	1,900	false
May 1, 2026 @ 20:09:20.556	192.168.1.37	5,353	224.0.0.251	5,353	false
May 1, 2026 @ 20:09:20.555	192.168.1.25	53,545	192.168.1.255	57,621	false
May 1, 2026 @ 20:09:20.555	192.168.1.25	53,546	192.168.1.255	57,621	false
May 1, 2026 @ 20:09:20.555	192.168.1.25	53,553	192.168.1.255	57,621	false
May 1, 2026 @ 20:09:20.555	192.168.1.25	53,554	192.168.1.255	57,621	false
May 1, 2026 @ 20:09:20.555	192.168.1.25	53,556	192.168.1.255	57,621	false
May 1, 2026 @ 20:09:20.555	-	-	-	-	-
May 1, 2026 @ 20:09:20.554	192.168.1.25	53,555	192.168.1.255	57,621	false
May 1, 2026 @ 20:09:20.554	192.168.1.25	53,540	192.168.1.255	57,621	false
May 1, 2026 @ 20:09:20.554	192.168.1.25	53,551	192.168.1.255	57,621	false
May 1, 2026 @ 20:09:20.554	192.168.1.25	53,541	192.168.1.255	57,621	false
May 1, 2026 @ 20:09:20.554	192.168.1.25	53,552	192.168.1.255	57,621	false

elastic | Search Elastic

Discover | Options | New | Open | Share | Inspect | Save

Search | KQL | Last 1 hour | Show dates | Refresh

+ Add filter

wazuh-alerts-*

Search field names

Filter by type 0

Selected fields 3
- agent.name
- agent.ip
- manager.name

Available fields 78
- _id
- _index
- _score
- _type
- @timestamp
- agent.ephemeral_id
- agent.hostname
- agent.id
- agent.type
- agent.version
- data.arch
- data.dpkg_status
- data.package
- data.version
- data.win.eventdata

340 hits | Chart options

May 1, 2026 @ 19:17:46.677 - May 1, 2026 @ 20:17:46.677

Time	agent.name	agent.ip	manager.name
May 1, 2026 @ 20:17:07.037	DC-1	10.0.2.100	wazuh.manager
May 1, 2026 @ 20:17:07.031	DC-1	10.0.2.100	wazuh.manager
May 1, 2026 @ 20:16:52.763	DC-1	10.0.2.100	wazuh.manager
May 1, 2026 @ 20:16:52.740	DC-1	10.0.2.100	wazuh.manager
May 1, 2026 @ 20:16:52.687	DC-1	10.0.2.100	wazuh.manager
May 1, 2026 @ 20:16:52.666	DC-1	10.0.2.100	wazuh.manager
May 1, 2026 @ 20:16:33.925	DC-1	10.0.2.100	wazuh.manager
May 1, 2026 @ 20:16:23.353	DC-1	10.0.2.100	wazuh.manager
May 1, 2026 @ 20:16:23.323	DC-1	10.0.2.100	wazuh.manager
May 1, 2026 @ 20:16:23.308	DC-1	10.0.2.100	wazuh.manager
May 1, 2026 @ 20:16:23.275	DC-1	10.0.2.100	wazuh.manager
May 1, 2026 @ 20:16:23.259	DC-1	10.0.2.100	wazuh.manager
May 1, 2026 @ 20:16:23.244	DC-1	10.0.2.100	wazuh.manager

10. Detection Capabilities

Network Layer (Suricata + pfSense)

Threat Type	Detection Method	Example Alert
Port Scanning	Threshold detection	"ET SCAN Nmap scan"
Malware C2	Signature matching	"ET MALWARE C2 Channel"
Suspicious Protocols	Protocol anomaly	"ET INFO Suspicious User-Agent" "DOS Slowloris Attack"
DDoS Patterns	Rate-based detection	
Known Exploits	Vulnerability signatures	"ET EXPLOIT EternalBlue"

Endpoint Layer (Wazuh)

Threat Type	Detection Method	Example Alert
Malware Execution	File integrity + signatures	"Malware detected"
Privilege Escalation	Auditd monitoring Registry monitoring	"sudo abuse detected"
Persistence	Authentication logs	"Startup entry added"
Lateral Movement		"Multiple failed logins"

Firewall Layer (pfSense Logs)

Threat Type	Detection Method	Example Alert
Unauthorized Access	Blocked connection logs	"Blocked inbound connection"
Reconnaissance	Multiple blocked ports	"Port scan detected"
Policy Violations	Rule match violations	"Invalid protocol"
VPN Issues	VPN Authentication logs	"VPN auth failure"

11. Attack Simulations & Validation

```
┌──(kali㉿kali)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.1.112  netmask 255.255.255.0  broadcast 192.168.1.255
        inet6 fe80::6c49:e9fb:1fb6:29f4  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:1f:b7:23  txqueuelen 1000  (Ethernet)
        RX packets 704  bytes 56464 (55.1 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 29  bytes 6112 (5.9 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

IIP address of the attacker → 192.168.1.112

```
┌──(kali㉿kali)-[~]
└─$ # From Kali - Scan pfSense WAN interface
nmap -sS 192.168.1.40
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-01 13:35 EDT
Nmap scan report for 192.168.1.40 (192.168.1.40)
Host is up (0.00082s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT     STATE SERVICE
22/tcp   open  ssh
53/tcp   open  domain
2222/tcp open  EtherNetIP-1
8443/tcp open  https-alt
MAC Address: 08:00:27:B8:FC:4F (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 4.63 seconds

┌──(kali㉿kali)-[~]
└─$ arp -a
? (192.168.1.40) at 08:00:27:b8:fc:4f [ether] on eth0
? (192.168.1.25) at d8:43:ae:1e:ea:4f [ether] on eth0
? (192.168.1.39) at 2c:cf:67:74:4d:07 [ether] on eth0
? (192.168.1.1) at c4:27:28:45:dd:dc [ether] on eth0

┌──(kali㉿kali)-[~]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ curl -k https://192.168.1.40:8443
curl: (35) TLS connect error: error:0A00010B:SSL routines::wrong version number

┌──(kali㉿kali)-[~]
└─$ curl -k https://192.168.1.41:8443
curl: (35) TLS connect error: error:0A00010B:SSL routines::wrong version number

┌──(kali㉿kali)-[~]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ sudo ping -f 192.168.1.40 -c 500
[sudo] password for kali:
PING 192.168.1.40 (192.168.1.40) 56(84) bytes of data.

--- 192.168.1.40 ping statistics ---
500 packets transmitted, 500 received, 0% packet loss, time 119ms
rtt min/avg/max/mdev = 0.071/0.193/10.560/0.482 ms, ipg/ewma 0.238/0.399 ms
```

```
┌──(kali㉿kali)-[~]
└─$ nmap -A 192.168.1.39
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-01 13:45 EDT
Nmap scan report for 192.168.1.39 (192.168.1.39)
Host is up (0.0027s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
22/tcp   open  ssh         OpenSSH 9.2p1 Debian 2+deb12u5 (protocol 2.0)
| ssh-hostkey:
|   256 12:3f:d1:2c:75:f7:9c:5f:74:b2:4a:83:c9:5a:82:df (ECDSA)
|_  256 ba:d8:7a:f2:3c:10:27:61:c2:2c:99:dd:5c:bc:9e:12 (ED25519)
139/tcp  open  netbios-ssn Samba smbd 4
445/tcp  open  netbios-ssn Samba smbd 4
5900/tcp open  vnc         VNC (protocol 3.8)
| vnc-info:
|   Protocol version: 3.8
|   Security types:
|     VeNCrypt (19)
|     Unknown security type (129)
|     RA2 (5)
|   VeNCrypt auth subtypes:
|_    Plain, Server-authenticated TLS (262)
9100/tcp open  jetdirect?
MAC Address: 2C:CF:67:74:4D:07 (Raspberry Pi (Trading))
Device type: general purpose|router
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_nbstat: NetBIOS name: RASPBERRYPI, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
| smb2-time:
|   date: 2026-05-01T17:45:31
|_  start_date: N/A
|_clock-skew: 1s
| smb2-security-mode:
|   3:1:1:
|_    Message signing enabled but not required

TRACEROUTE
HOP RTT     ADDRESS
1   2.69 ms 192.168.1.39 (192.168.1.39)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 38.63 seconds
```

elastic

Search Elastic

Discover | Options | New | Open | Share | Inspect | Save

src_ip : 192.168.1.112 AND flow.alerted : true | KQL | Last 15 minutes | Show dates | Refresh

+ Add filter

suricata-*

Search field names

Filter by type 0

Selected fields 3

- src_ip
- dest_ip
- flow.alerted

Available fields 610

Popular

- alert.action
- alert.category
- alert.signature
- dest_port
- _id
- _index
- _score
- _type
- @timestamp
- agent.ephemeral_id
- agent.hostname
- agent.id
- agent.name

8 hits

Chart options

May 1, 2026 @ 20:33:59.795 - May 1, 2026 @ 20:48:59.795

Time	src_ip	dest_ip	flow.alerted
May 1, 2026 @ 20:46:51.258	192.168.1.112	192.168.1.39	true
May 1, 2026 @ 20:46:50.256	192.168.1.112	192.168.1.39	true
May 1, 2026 @ 20:46:40.242	192.168.1.112	192.168.1.39	true
May 1, 2026 @ 20:46:39.241	192.168.1.112	192.168.1.39	true
May 1, 2026 @ 20:46:39.241	192.168.1.112	192.168.1.39	true
May 1, 2026 @ 20:46:38.240	192.168.1.112	192.168.1.39	true
May 1, 2026 @ 20:46:37.239	192.168.1.112	192.168.1.39	true
May 1, 2026 @ 20:46:36.238	192.168.1.112	192.168.1.39	true

	0/0 B	IPv4 TCP	192.168.1.112	*	10.0.3.13	22 (SSH)	*	none	BLOCK ATTACKER

```
┌──(kali㉿kali)-[~]
└─$ hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://192.168.1.41:2222

Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-01 13:40:32
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per task
[DATA] attacking ssh://192.168.1.41:2222/
[ERROR] could not connect to ssh://192.168.1.41:2222 - Timeout connecting to 192.168.1.41
```

elastic

Search Elastic

Discover | Options | New | Open | Share | Inspect | Save

src_ip : 192.168.1.112 | KQL | Last 15 minutes | Show dates | Refresh

+ Add filter

pfsense-*

Search field names

Filter by type 0

Selected fields 3

- dest_ip
- dest_port
- action

Available fields 20

Popular

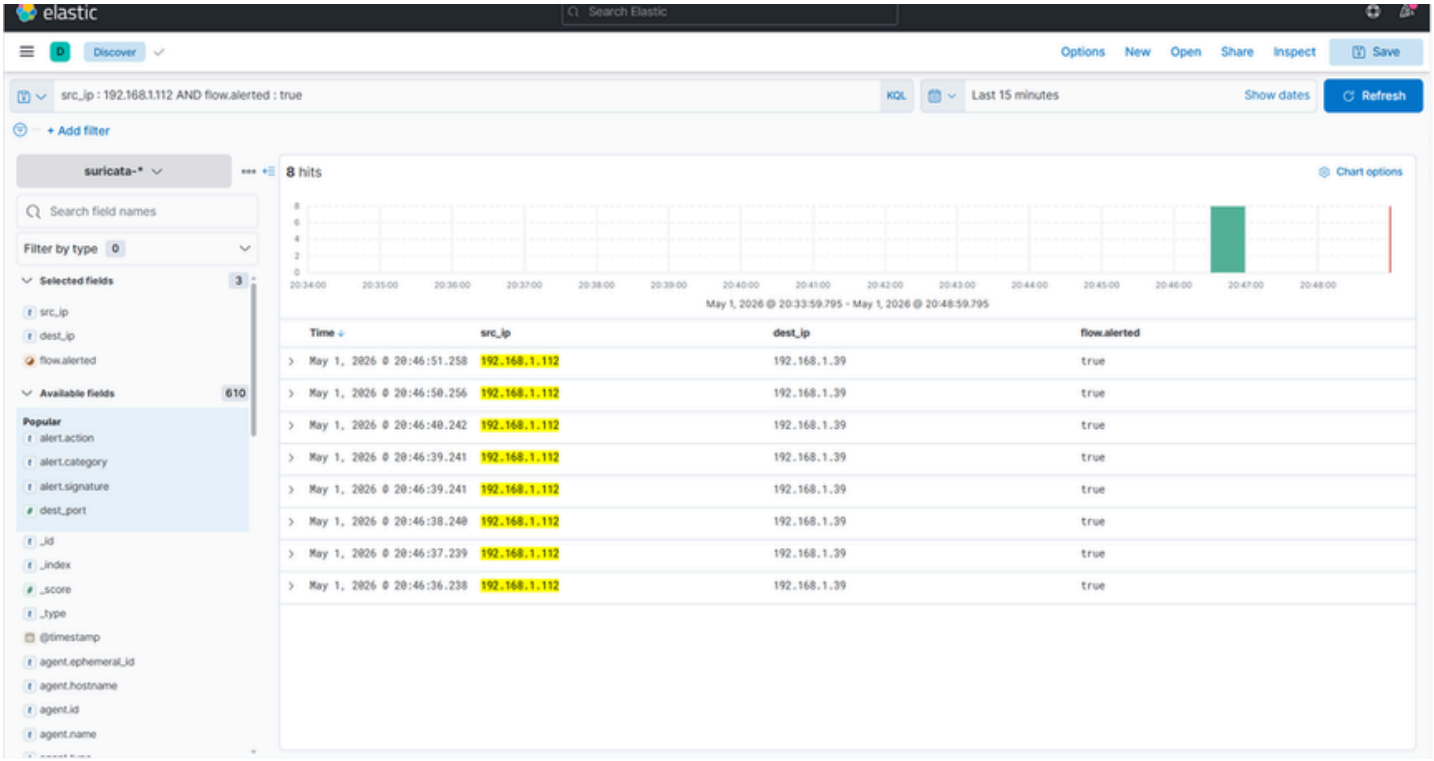
- src_ip
- src_port
- _id
- _index
- _score
- _type
- @timestamp
- @version
- column23
- column24
- column25
- column26
- column27

1,702 hits

Chart options

May 1, 2026 @ 20:25:38.398 - May 1, 2026 @ 20:40:38.398

Time	dest_ip	dest_port	action
May 1, 2026 @ 20:40:15.591	192.168.1.41	52,806	block
May 1, 2026 @ 20:40:15.591	192.168.1.41	52,860	block
May 1, 2026 @ 20:40:15.591	192.168.1.41	52,882	block
May 1, 2026 @ 20:40:15.591	192.168.1.41	52,844	block
May 1, 2026 @ 20:40:15.591	192.168.1.41	52,762	block
May 1, 2026 @ 20:40:15.591	192.168.1.41	52,790	block
May 1, 2026 @ 20:40:15.591	192.168.1.41	52,898	block
May 1, 2026 @ 20:40:15.591	192.168.1.41	52,868	block
May 1, 2026 @ 20:40:15.591	192.168.1.41	52,750	block
May 1, 2026 @ 20:40:15.591	192.168.1.41	52,826	block
May 1, 2026 @ 20:40:15.591	192.168.1.41	52,798	block
May 1, 2026 @ 20:40:15.591	192.168.1.41	52,812	block
May 1, 2026 @ 20:40:15.591	192.168.1.41	52,774	block




```
┌──(kali㉿kali)-[~]
└─$ enum4linux 192.168.1.39 -a
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Fri May  1 13:57:30 2026

 ========================== ( Target Information ) ==========================

Target ........... 192.168.1.39
RID Range ........ 500-550,1000-1050
Username ......... ''
Password ......... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none


 ============ ( Enumerating Workgroup/Domain on 192.168.1.39 ) ============


[+] Got domain/workgroup name: WORKGROUP


 ============ ( Nbtstat Information for 192.168.1.39 ) ============

Looking up status of 192.168.1.39
        RASPBERRYPI     <00> -         B <ACTIVE>  Workstation Service
        RASPBERRYPI     <03> -         B <ACTIVE>  Messenger Service
        RASPBERRYPI     <20> -         B <ACTIVE>  File Server Service
        ..__MSBROWSE__. <01> - <GROUP> B <ACTIVE>  Master Browser
        WORKGROUP       <00> - <GROUP> B <ACTIVE>  Domain/Workgroup Name
        WORKGROUP       <1d> -         B <ACTIVE>  Master Browser
        WORKGROUP       <1e> - <GROUP> B <ACTIVE>  Browser Service Elections

        MAC Address = 00-00-00-00-00-00

 ============ ( Session Check on 192.168.1.39 ) ============


[+] Server 192.168.1.39 allows sessions using username '', password ''


 ============ ( Getting domain SID for 192.168.1.39 ) ============

Domain Name: WORKGROUP
Domain Sid: (NULL SID)

[+] Can't determine if host is part of domain or part of a workgroup


 ============ ( OS information on 192.168.1.39 ) ============


[E] Can't get OS info with smbclient


[+] Got OS info for 192.168.1.39 from srvinfo:
        RASPBERRYPI    Wk Sv PrQ Unx NT SNT Samba 4.17.12-Debian
        platform_id     :       500
        os version      :       6.1
        server type     :       0x809a03
```

elastic

Search Elastic

Discover

Options New Open Share Inspect Save

src_ip : 192.168.1.112 AND flow.alerted : true

KQL

Last 15 minutes

Show dates

Refresh

+ Add filter

suricata-*

Search field names

Filter by type 0

Selected fields 4

- src_ip
- dest_ip
- dest_port
- flow.alerted

Available fields 638

Popular

- alert.action
- alert.category
- alert.signature
- _id
- _index
- _score
- _type
- @timestamp
- agent.ephemeral_id
- agent.hostname
- agent.id
- agent.name

12 hits

Chart options

May 1, 2026 @ 20:44:44.821 - May 1, 2026 @ 20:59:44.821

Time	src_ip	dest_ip	dest_port	flow.alerted
May 1, 2026 @ 20:58:42.054	192.168.1.112	192.168.1.39	139	true
May 1, 2026 @ 20:58:36.046	192.168.1.112	192.168.1.39	445	true
May 1, 2026 @ 20:58:36.046	192.168.1.112	192.168.1.39	139	true
May 1, 2026 @ 20:50:35.415	192.168.1.112	192.168.1.39	-	true
May 1, 2026 @ 20:46:51.258	192.168.1.112	192.168.1.39	5,900	true
May 1, 2026 @ 20:46:50.256	192.168.1.112	192.168.1.39	5,900	true
May 1, 2026 @ 20:46:40.242	192.168.1.112	192.168.1.39	445	true
May 1, 2026 @ 20:46:39.241	192.168.1.112	192.168.1.39	445	true
May 1, 2026 @ 20:46:39.241	192.168.1.112	192.168.1.39	445	true
May 1, 2026 @ 20:46:38.240	192.168.1.112	192.168.1.39	445	true
May 1, 2026 @ 20:46:37.239	192.168.1.112	192.168.1.39	445	true
May 1, 2026 @ 20:46:36.238	192.168.1.112	192.168.1.39	5,900	true

12. Maintenance & Troubleshooting

Daily Health Checks

```
# Check all Docker containers
docker ps

# Check Elasticsearch health
curl -s "http://192.168.1.55:9200/_cluster/health" | jq .

# Check recent logs in Kibana (last 15 minutes)
# Verify data is present
```

Weekly Maintenance

Task	Command
Clear old indices	`curl -X DELETE "http://192.168.1.55:9200/pfsense-2025.03.*"`
Update Suricata rules	`docker exec suricata suricata-update`
Verify disk space	`df -h`
Backup configurations	Export from pfSense and Kibana

Common Issues & Solutions

Issue	Symptom	Solution
Logstash not receiving	No pfsense-* index	Check pfSense remote syslog settings
Suricata no alerts	No suricata-* data	Check suricata -i eth0 interface
Wazuh agents disconnected	No wazuh-alerts-*	`systemctl restart wazuh-agent`
Kibana no data	"No results found"	Check time range, index pattern
High memory usage	Slow queries	Increase ES_JAVA_OPTS memory

Log Rotation Policy

Index	Retention	Cleanup Command
pfsense-*	30 days	Delete older indices
suricata-*	30 days	Delete older indices
wazuh-alerts-*	90 days	Delete older indices
wazuh-archives-*	7 days	Delete older indices

Backup Strategy

```
# Backup Elasticsearch indices
docker exec elasticsearch elasticsearch-dump \
--input=http://localhost:9200/pfsense-* \
--output=/backup/pfsense-backup.json

# Backup Kibana dashboards
curl -X GET "http://localhost:5601/api/saved_objects/_export" \
-H "kbn-xsrf: true" \
 -d '{"type":["dashboard","visualization","index-pattern"]}' \
> kibana-backup.ndjson
```

Appendix A: Quick Reference Commands

Docker Management

```
# View all running containers
docker ps

# View container logs
docker logs [container_name] --tail 50

# Restart container
docker restart [container_name]
```

Elasticsearch Queries

```
# View all indices
curl -s "http://192.168.1.55:9200/_cat/indices?v"

# Search pfsense logs
curl -s "http://192.168.1.55:9200/pfsense-*/_search?size=5&sort=@timestamp:desc" | jq .

# Delete old indices
curl -X DELETE "http://192.168.1.55:9200/pfsense-2025.03.*"
```

Logstash Testing

```
# Check Logstash logs
docker logs logstash --tail 20

# Send test syslog message
echo "<134>Apr 30 18:36:36 filterlog[64922]: test" | nc -u 192.168.1.55 5140
```

Suricata Commands (Raspberry Pi)

```
# Check Suricata is running
docker exec suricata ps aux | grep suricata

# View recent alerts
docker exec suricata tail -20 /var/log/suricata/eve.json | jq 'select(.event_type=="alert")'

# Update rules
docker exec suricata suricata-update
```

Appendix B: Port Reference

Port	Service	Protocol	Purpose
22	SSH	TCP	Remote management
53	DNS	TCP/UDP	Name resolution
80	HTTP	TCP	Web services
443	HTTPS	TCP	Kibanai/Uptime Kuma
500	IKE	UDP	IPsec key exchange
1514	Wazuh	TCP	Agent communication Agent enrollment
1515	Wazuh	TCP	VPN through NAT
4500	IPsec NAT-T	UDP	pfSense log collection
5140	Syslog	UDP/TCP	Web dashboard
5601	Kibana	HTTP	Firewall management API queries
8443	pfSense GUI	TCP	
9200	Elasticsearch	HTTP	
3001	Uptime Kuma	HTTP	Service monitoring

Appendix C: MITRE ATT&CK Mapping

Detection	MITRE Tactic	MITRE Technique
Suricata alert	Initial Access	T1190 Exploit Public-Facing Application
Port scan detection	Reconnaissance	T1046 Network Service Scanning
Brute force detection	Credential Access	T1110 Brute Force
Malware alert	Execution	T1204 User Execution
Blocked firewall	Defense Evasion	T1562 Impair Defenses
C2 traffic detection	Command & Control	T1071 Application Layer Protocol